

RAGAS Evaluation

clear mental model for checking RAG answers



**RAG gives the answer.
RAGAS diagnoses the answer.**

We will check only 3 relationships:

- 1. $q \leftrightarrow c$ **Context Relevance** → retrieval quality
- 2. $c \leftrightarrow a$ **Faithfulness** → hallucination check
- 3. $a \leftrightarrow q$ **Answer Relevance** → user intent match

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q: User Question

Who directed Oppenheimer?

c: Retrieved Context

Oppenheimer was directed by Christopher Nolan.

a: Generated Answer

Christopher Nolan directed Oppenheimer.

Step 1 — Input Triplet

RAGAS starts with these 3 values:

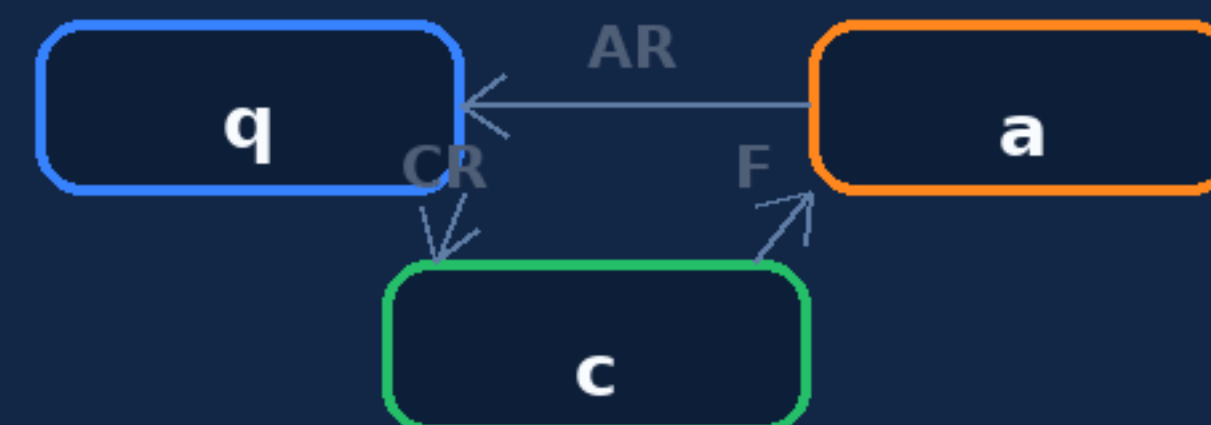
q = the user question

c = retrieved context/chunks

a = generated answer from LLM

Input to RAGAS
(q, c, a)

Mental model



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q: User Question

Who directed Oppenheimer?

c: Retrieved Context

Oppenheimer was directed by Christopher Nolan.

a: Generated Answer

Christopher Nolan directed Oppenheimer.

Step 2 — Context Relevance: q ↔ c

Question: Did retrieval bring useful and focused chunks?

RAGAS asks LLM:

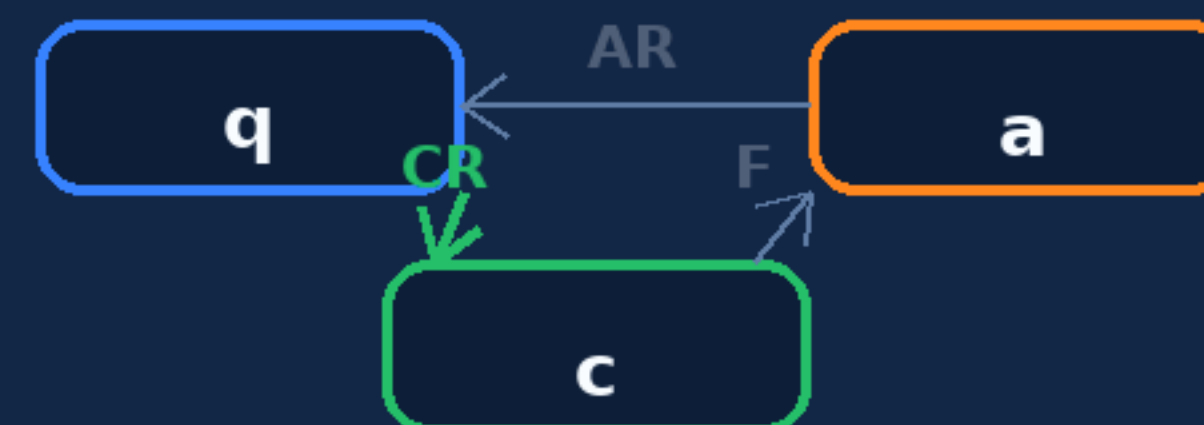
“Extract only sentences from context that help answer the question.”

Useful sentence: “Oppenheimer was directed by Christopher Nolan.”

CR =
useful sentences
—————
total sentences

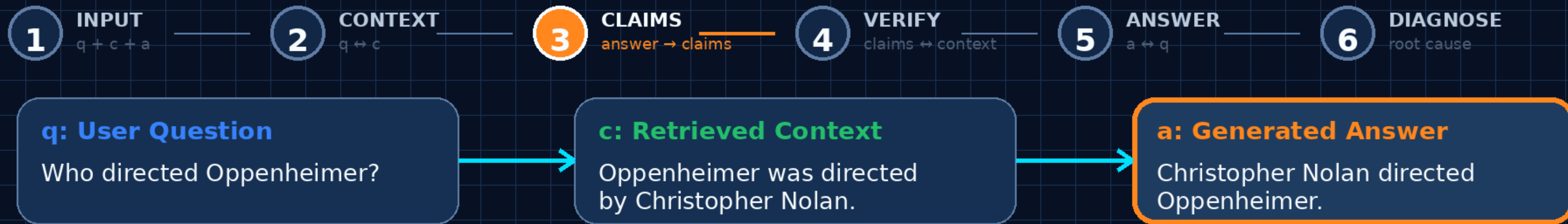
Low CR = noisy retrieval

Mental model



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Step 3 — Faithfulness Part A: break answer into claims

RAGAS first splits the answer into small factual statements.

Answer

Christopher Nolan directed Oppenheimer.

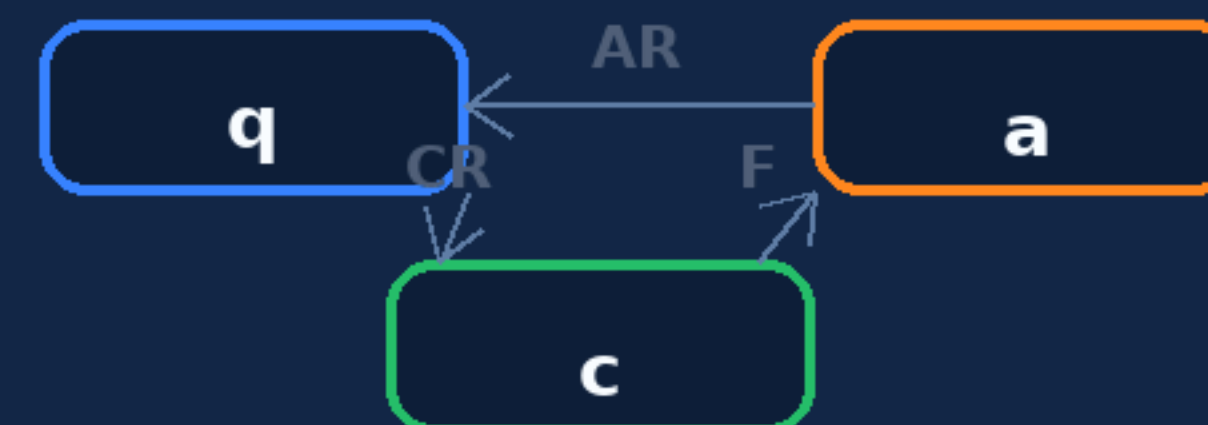
Atomic claims

S1: Christopher Nolan is the director.

S2: Christopher Nolan directed Oppenheimer.

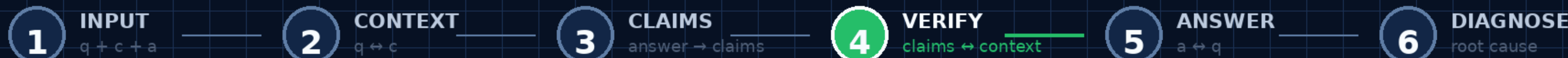
Now each claim can be verified.

Mental model



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q: User Question

Who directed Oppenheimer?

c: Retrieved Context

Oppenheimer was directed by Christopher Nolan.

a: Generated Answer

Christopher Nolan directed Oppenheimer.

Step 4 — Faithfulness Part B: verify claims against context

Each claim must be supported by the retrieved context.

S1: Christopher Nolan is the director.

S2: Christopher Nolan directed Oppenheimer.

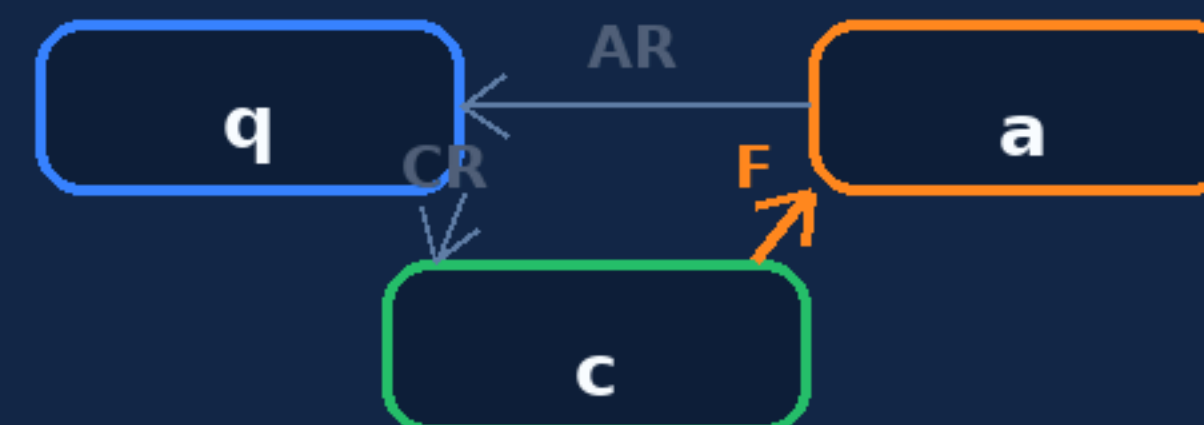
S1 → YES

S2 → YES

F = 2 / 2 = 1.0

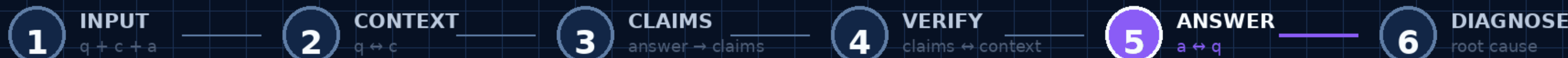
If any claim is not in context → faithfulness drops.

Mental model



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q: User Question

Who directed Oppenheimer?

c: Retrieved Context

Oppenheimer was directed by Christopher Nolan.

a: Generated Answer

Christopher Nolan directed Oppenheimer.

Step 5 — Answer Relevance: $a \leftrightarrow q$

Question: Does the answer match what the user asked?

Answer a

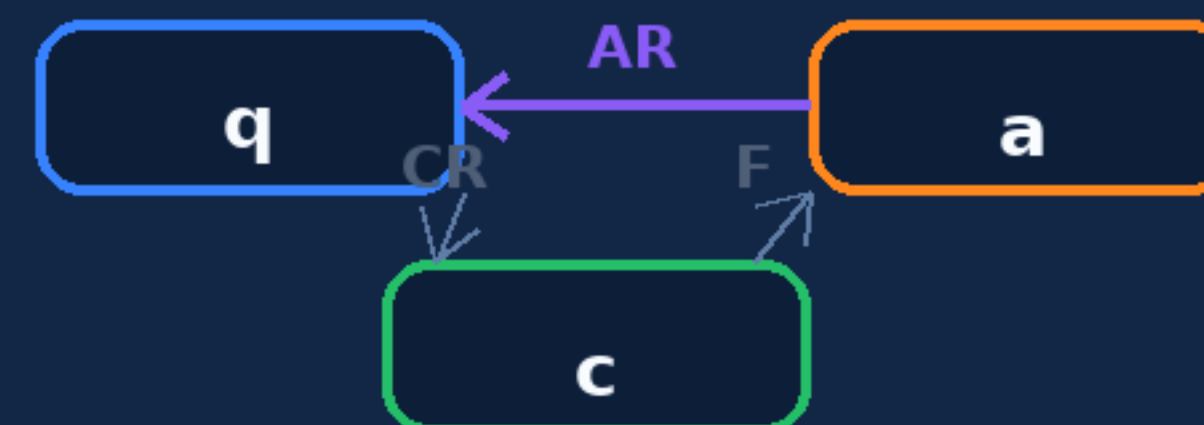
Christopher Nolan directed Oppenheimer.

Generate reverse question

q₁: Who directed Oppenheimer?

Then compare original q with q₁ using embeddings + cosine similarity.

Mental model



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q: User Question

Who directed Oppenheimer?

c: Retrieved Context

Oppenheimer was directed by Christopher Nolan.

a: Generated Answer

Christopher Nolan directed Oppenheimer.

Step 6 — Final Diagnosis

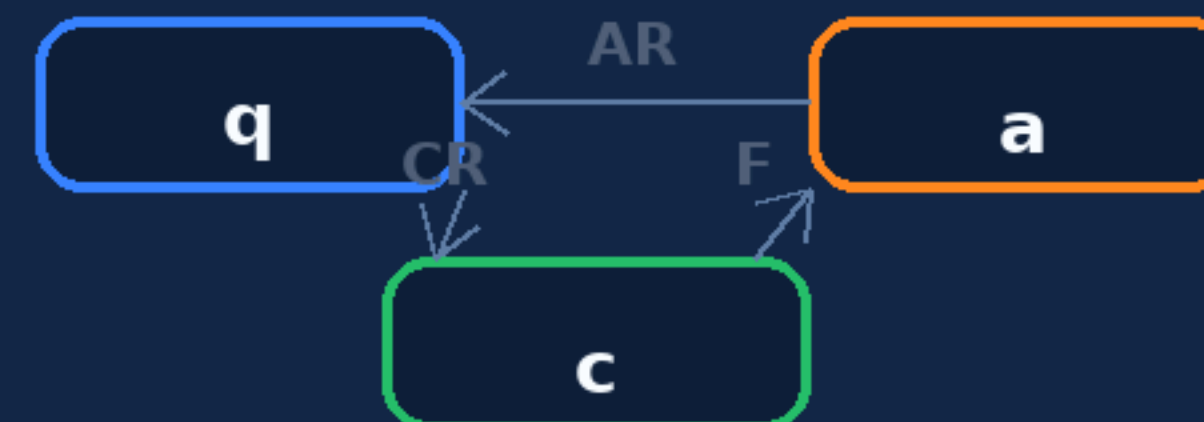
Use the low score to identify the weak part.

Low Context Relevance retriever brought wrong/noisy chunks
Fix: chunking, top-k, reranker, hybrid search

Low Faithfulness answer not grounded in context
Fix: prompt, citations, temperature, stronger model

Low Answer Relevance answer does not match user intent
Fix: query rewrite, direct-answer prompt

Mental model



Final memory hook

Context Relevance = $q \leftrightarrow c$ = retrieval quality

Faithfulness = $c \leftrightarrow a$ = grounding / hallucination

Answer Relevance = $a \leftrightarrow q$ = user intent match

RAGAS does not replace human review, but it gives fast debugging signals.